Notes on "ALMOST SHARP LOWER BOUND FOR THE NODAL VOLUME OF HARMONIC FUNCTIONS" by Prof. Andrea Sartori, Prof. Alexander Logunov, and Prof. Lakshmi Priya

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1 Why I'm reading this

I am interested in analysis; I'm exploring various subfields in analysis – I got to know about fields like Harmonic Analysis, Functional Analysis, PDEs, and other broader fields like Geometric Measure Theory via this paper.

2 Prerequisites

- Laplacian of a function
- Nodal set of a function f
- Harmonic functions
- Diameter of a set – largest (supremum) distance between two points in the set (under a metric space X where the metric d is clearly defined)
- s-dimensional Hausdorff measure of a set in any metric space (particularly $\mathbb{R}^n$ for this paper.
- Hausdorff dimension – measures the critical point at which the Hausdorff measure of the set transitions between 0 and ∞. For the set $\mathbb{R}^n$, the Hausdorff dimension is $\dim_H(\mathbb{R}^n) = n$
- Harnack's Inequality for harmonic functions
- Maximum principle
- Convex sets

3 Basic definitions

- Harmonic Functions

  A function $u : \mathbb{R}^n \to \mathbb{R}$ is called harmonic if its Laplacian $\Delta u = 0$, where

  $$\Delta u = (\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2} + \cdots + \frac{\partial^2}{\partial x_n^2})u = \sum_{k=1}^{n} \frac{\partial^2}{\partial x_k^2} u(x_1, x_2, \ldots, x_n)$$

- s-dimensional Hausdorff measure of a set

  The $s-$ dimensional Hausdorff measure of a set E in metric space X is defined by

  $$\mathcal{H}^s(E) = \lim_{\delta \to 0} H_\delta^s(E)$$

  where H_δ^s is defined by

  $$H_\delta^s(E) = \inf_{E \subset \bigcup_i U_i, \mathrm{diam}(U_i) \leq \delta} \sum_i \mathrm{diam}(U_i)^s$$

  In essence, we cover E with a countable collection $\{U_i \in X\}$ and compute $\sum_i \mathrm{diam}(U_i)^s$ for that collection. Then, taking the smallest, i.e., the infimum over all these sums, we get the Hausdorff measure at scale δ. Taking the limit as $\delta \to 0$, that is, by considering sets with very small diameter, gives us the Hausdorff measure – which is just a fancy term that provides a notion of $s-$dimensional size for the set. For example, for $s = 1$, the Hausdorff measure corresponds to the length of the set E. Suppose $E = [0, 1]$; Then, $\mathcal{H}^1(E) = 1$.

- Doubling Index of h in B

  The doubling index of h in ball B gives us an estimate of the ratio of the highest magnitude of h in 2B to that in B. It is defined by

  $$N_h(B(x, r)) = \log_2 \frac{\sup_{2B} |h|}{\sup_B |h|}$$

- Scaled Doubling index of h in B

  To account for the geometrical significance of the ball itself, we use the Scaled Doubling Index defined by

  $$SN(B) := N(B)\mathrm{radius}(B)^{n-1}$$

- Maximum Doubling Index of a cube Q

  We denote the Maximum Doubling Index of a cube $Q \in \mathbb{R}^n$ by

  $$N^*(Q) = \sup_{x \in Q, 0 < \rho \leq \text{Diam}\{Q\}} N(B(x, \rho))$$

  As the name suggests, this gives us the supremum of all the doubling index considering balls with different centers $x \in Q$ and radii ρ, where ρ ranges from $0 < \rho \leq \text{Diam}(Q)$

- Bounded Doubling Index in ball B

  For a harmonic function, $N_u(B)$ is said to be bounded if there exists a constant $C > 0$ such that

  $$N_u(B) = N_u(B(x, r)) \leq C$$

- Nodal sets

  The nodal set of a harmonic function $u : \mathbb{R}^n \to \mathbb{R}$ denoted by $\{u = 0\}$ are precisely the set of all $x \in \mathbb{R}^n$ such that,

  $$u(x) = 0$$

- Nodal volume

  The nodal volume simply refers to the Hausdorff measure of the nodal set of a harmonic function u.

- Gradient of a harmonic function u

  The gradient of a harmonic function $u : U \subset \mathbb{R}^n \to \mathbb{R}$ is defined as the vector that points in the direction of steepest increase of u. It is computed by taking the partial derivatives

  $$\nabla u(x) = (\frac{\partial u}{\partial x_1}, \ldots, \frac{\partial u}{\partial x_n})$$

  at each $x \in U$

- Homothetic copy of cube $Q \in \mathbb{R}^n$, and Homothety Coefficient λ

  A homothetic copy of a cube $Q \in \mathbb{R}^n$ is simply obtained by scaling Q uniformly by a factor called the Homethety coefficient λ. The new cube obtained will have side length λs where s is the side length of the original cube Q.

- Frequency function

  For a given harmonic function $u : \mathbb{R}^n \to \mathbb{R}$, define $H(x, r)$ by

  $$H(x, r) = \int_{\partial B(x,r)} u^2 dS$$

  where dS is the surface measure on $\partial B(x, r)$, $r > 0$, $x \in \mathbb{R}^n$

Let's understand this intuitively: Consider tiny patches on the sphere of area dS. Now, in one of the points on this path, calculate the value of u. Integrating $u^2 dS$ over all the patches gives us $H(x,r)$ which can be thought of as the 'size' of the function u on the boundary of sphere $B(x,r)$.

Also define

$$G(x,r) = \int_{B(x,r)} |\nabla u|^2 dV$$

where dV is the volume measure on $B(x,r)$.

To understand this intuitively: Consider splitting the ball $B(x,r)$ into infinitely many small volumes dV. In each of these small volumes, we consider the product $|\nabla u|^2 dV$ – which measures how rapidly u is changing locally, squared. $G(x,r)$ can be thought of as the total amount of variation of u throughout the ball. We can also think of G as energy associated with u. $\int |\nabla u|^2$ is sometimes called as **Dirichlet Energy**.

Finally, we define the Frequency Function as

$$\beta(x,r) = \frac{rG(x,r)}{H(x,r)}$$

which can be thought of as the energy (G) being distributed across the size (H) of the harmonic function u in the ball B, scaled by a factor of r. Equivalently,

$$\beta(x,r) := \frac{rH'(x,r)}{H(x,r)} - \frac{n-1}{2}$$

where the derivative is taken with respect to r. This form can be thought of as the logarithmic derivative of H, scaled by factor r with the constant $\frac{n-1}{2}$

4 Theorems, Lemmas and Claims

We will now give a very vague proof outline of some of the listed claims, lemmas, and theorems. All of which are listed below have only been defined informally. Note that proofs from the Appendix section of the paper have not been considered yet.

Theorem 1 *(Main) The main theorem states that for a unit ball B in an $n-$dimensional Euclidean space where $n \geq 3$, the Hausdorff measure in dimension (n-1) of the nodal set of the harmonic function $u : 4B \to \mathbb{R}$ in the ball $2B$ is **bounded below** by the doubling index of $\frac{1}{2}B$ to the power $1-\varepsilon$. Equivalently,*

$$\mathcal{H}^{n-1}(\{u = 0\} \cap 2B) \geq N(\frac{1}{2}B)^{1-\varepsilon}$$

Theorem 2 *Let $B \in \mathbb{R}^n$ be a unit ball. Then, the Hausdorff measure in dimension $n-1$ of the zero set of u in B is* ***bounded above*** *by a constant multiple of the doubling index $N(B)$ for all harmonic functions $u : 2B \to \mathbb{R}$*

Claim 1 *Let $B = B(x,r) \subset \mathbb{R}^n$ be any ball. Then for any non-constant harmonic function u defined in $2B$ satisfying $u(x) \geq 0$, we have*

$$\sup_{\frac{4}{3}B} u \gtrsim_n \sup_B |u|$$

Proof outline of Claim 1: Let $M = \sup_B u$ and $m = \inf_B u$.

- If $m \geq 0$, then $u \geq 0$ and so $|u| = u$. It is well known that for $A \subset B$

  $$\sup_A u \leq \sup_B u$$

  Hence, the conclusion follows.

- If $m < 0$, then it suffices to show that $-m \lesssim_n M'$ where $M' = \sup_{4/3B} u$: Since $m < 0$, there exists points $x_n \neq 0$ for which $u(x_n) < 0$. For these points, $|u| = -u$. We know that

  $$\sup_B |u| = \sup_B(-u) = -\inf_B u = -m$$

  Let $x_m, x_M \in \partial B$ be such that $u(x_m) = m$ and $u(x_M) = M$ – these values are attainable by the Extreme Value Theorem which applies for compact sets such as $\overline{B}$. Now, by maximum principle, the extrema m and M occur on the boundary. Then, we define $h = M' - u$ in $\frac{4}{3}B$. By Harnack's inequality (not mentioned yet), there is a constant $C = C(n) > 1$ such that

  $$M' - m = h(x_m) \leq Ch(x_M) \leq C(M' - M) \leq CM'$$

  Hence,

  $$-m \lesssim_n M'$$

Claim 2 *Let $f : D \to \mathbb{R}$ be a continuous function in a* ***convex set*** *$D \subset \mathbb{R}^n$. Assume that there are two balls B^+ and B^- both of radius $r > 0$ such that $f > 0$ in B^+ and $f < 0$ in B^-. Then*

$$\mathcal{H}^{n-1}(\{f = 0\}) \gtrsim_n r^{n-1}$$

Proof outline of Claim 2:
Let B^+ be the ball such that $f > 0$ inside. Similarly, let B^- be the ball such that $f < 0$ inside. Join the points in the two balls by line segments parallel to the line segment joining the centers of B^+ and B^-. Since

f is positive at one end and negative at the other end, every such line segment must contain a zero of f. Why? – By an application of IVT? Consider a hyperplane orthogonal to these line segments such that $f = 0$ for at least one point on the plane. Now, project the zero set $\{f = 0\}$ onto the hyperplane. These zeroes of f in the line segments form a $(n-1)$ dimensional ball of radius r (same as that of B^+ and B^-).

We take advantage of the fact that the orthogonal projection does not increase the distances; Therefore, the Hausdorff measure of $\{f = 0\}$ is at least the Hausdorff measure of its projection which contains the $n-1$ dimensional ball of radius r. This has Hausdorff measure in the order of r^{n-1}. This is an interesting technique to find the lower bound – essentially finding an invariant while simultaneously reducing/simplifying the problem.

Together with Claims 1 and 2, we are able to prove the next lemma:

Lemma 1 *Let $B = B(0,1) \subset \mathbb{R}^n$ be a unit ball and let u be a harmonic function in $2B$. Suppose that $u(0) = 0$, or in other words, $0 \in \{u = 0\}$, then $\exists$ constant $c = c(n) > 0$ s.t.*

$$\mathcal{H}^{n-1}(\{u = 0\} \cap 2B) \geq cN(B)^{1-n}$$

Proof outline of Lemma 1:
We will consider the Hausdorff measure of the zero set of u in the spherical layer $\{4/3 \leq |x| \leq 5/3\}$. We define $N = 20N(B)$. We define the concentric spheres

$$S_i = \partial(r_i B) = \{x \in 2B : |x| = r_i\} \text{ where } r_i = \frac{4}{3} + \frac{i}{3N}$$

for $i = 0, 1, \ldots, N$. Notice that there are $N+1$ such spherical fragments. We will now consider the maximum and minimum value that u takes in these fragments. Let

$$m_i^+ := \max_{S_i} u \qquad m_i^- := \min_{S_i} u$$

By assumption, u is a non-constant harmonic function and therefore u is not identically zero. We have, by assumption, that $u(0) = 0$ or equivalently, the center of $B(r)$ for any $r > 0$ is an element in the nodal set. As discussed before, the maximum principle implies that the extrema cannot occur entirely on the interior. Hence,

$$m_i^+ > 0, \quad m_i^- < 0$$

and

$$m_i^+ < m_{i+1}^+, \quad |m_i^-| < |m_{i+1}^-|$$

With the help of Claim 2, we only have to show that there are two balls each of radius r in B, such that u is positive in one ball and negative in the other. Taking $r = c_0 N^{-1}$ in the claim for some $c_0 = c_0(n) > 0$, we obtain the desired lower bound.
The increment of u from $\frac{4}{3}B$ to $\frac{5}{3}B$ can be written as a telescopic product – what we really mean by "increment" is the multiplicative growth $\frac{m^+_{i+1}}{m^+_i}$. So the increment of u from $\frac{4}{3}B$ to $\frac{5}{3}B$ is

$$\frac{m^+_N}{m^+_0} = \frac{\sup_{\frac{5}{3}B} u}{\sup_{\frac{4}{3}B} u} = \frac{m^+_1}{m^+_0} \cdot \frac{m^+_2}{m^+_1} \cdot \ldots \frac{m^+_N}{m^+_{N-1}}$$

By Claim 1, $\sup_{\frac{4}{3}B} u \gtrsim \sup_B |u| \implies \frac{1}{\sup_{\frac{4}{3}B} u} \lesssim \frac{1}{\sup_B |u|}$. And, $\sup_{\frac{5}{3}B} u \leq \sup_{2B} |u|$. Thus,

$$\frac{\sup_{\frac{5}{3}B} u}{\sup_{\frac{4}{3}B} u} \leq \frac{\sup_{2B} |u|}{\sup_B |u|}$$

By the definition of doubling index $N(B)$,

$$\frac{\sup_{\frac{5}{3}B} u}{\sup_{\frac{4}{3}B} u} \leq \frac{\sup_{2B} |u|}{\sup_B |u|} = 2^{N(B)} = 2^{N/20} \leq e^N$$

Thus, since the product of the multiplicative growths are bounded above, each of them must be bounded above too (there must be finitely many indices for which the increments are bounded above):

$$m^+_{i+1}/m^+_i \leq C_1 \iff m^+_{i+1} \leq C_1 m^+_i$$

where $C_1 = C_1(n) > 0$ This holds true for at least $\frac{3}{4}N$ values of i – why? Similarly,

$$|m^-_{i+1}|/|m^-_i| \leq C_1 \iff |m^-_{i+1}| \leq C_1 |m^-_i|$$

From now we will be considering those values of i for which both satisfy.

Let x^+_i, x^-_i be the points such that $u(x^+_i) = m^+_i$ and $u(x^-_i) = m^-_i$. Define

$$B^+_i := B(x^+_i, \frac{1}{10N}) \quad \textit{and} \quad B^-_i := B(x^-_i, \frac{1}{10N})$$

We want to show that B^+_i has $u > 0$ and B^-_i has $u < 0$ – the choice of radius $\frac{1}{10N}$ was to choose a small enough ball such that this happens. By a standard gradient estimate, Claim 2 and the fact that $\frac{8}{3}B^+_i \subset r_{i+1}B$, we have that

$$\sup_{B^+_i} |\nabla u| \lesssim_n N m^+_i$$

That is, the gradient is bounded above. From this, we can conclude that for a small enough radius ($= \frac{1}{10N}$) there exists balls B such that $u > 0$ or $u < 0$ inside. That is, $\exists c_0 \in (0,1)$ such that $u > 0$ in $c_0 B_i^+$. Similarly,$\exists c_0 \in (0,1)$ such that $u < 0$ in $c_0 B_i^-$. This lets us invoke the result of Claim 2 to show that for $r = \frac{c_0}{10N}$

$$\begin{aligned} \mathcal{H}^{n-1}(\{f=0\} \cap 2B) \gtrsim_n (10c_0 N^{-1})^{n-1} &= (2c_0 N(B)^{-1})^{n-1} && (1) \\ &= 2^{n-1} c_0^{n-1} N(B)^{1-n} && (2) \\ &= C(n) N(B)^{1-n} && (3) \end{aligned}$$

or equivalently

$$\mathcal{H}^{n-1}(\{f=0\} \cap 2B) \geq c(n) N(B)^{1-n}$$

Note that Theorem 2 is not of much significance in this paper as it focuses on the upper bound, but it helps us in our next claim. Theorem 2 and Lemma 1 imply that the nodal volume of u is comparable to the surface area of B:

Claim 3 *Let $B \subset \mathbb{R}^n$ be a ball and let $C > 1$ be some constant. Then for all harmonic functions u in $2B$, vanishing at the center of B, such that $N_u(B) \leq C$ (i.e. bounded doubling index), we have*

$$\mathcal{H}^{n-1}(\{u=0\} \cap B) \asymp_{C,n} r^{n-1}$$

It is also comparable to the scaled doubling index:

$$\mathcal{H}^{n-1}(\{u=0\} \cap B) \asymp_{C,n} SN(B)$$

Theorem 3 *For any harmonic function in $\mathbb{R}^n$, its frequency function $\beta(r) := \beta(0,r)$ is a non-decreasing function of r.*

Corollary of Theorem 3:

For any harmonic function in $\mathbb{R}^n$, let β be defined as in Theorem 3. Let $0 < r_1 < r_2$. Then, we have

$$\frac{H(r_2)}{r_2^{n-1}} = \frac{H(r_1)}{r_1^{n-1}} e^{2\int_{r_1}^{r_2} \frac{\beta(r)}{r} dr}$$

Notice that, $H(r_1)/r_1^{n-1}$ and $H(r_2)/r_2^{n-1}$ has the same structure: Define $\alpha(r) = \frac{H(r)}{r^{n-1}}$. Then we have that,

$$\alpha(r_2) = \alpha(r_1) \exp(2 \int_{r_1}^{r_2} \frac{\beta(r)}{r} dr)$$

To have a sort of a vague understanding of what's happening here (as far as I could understand): The functional value of α at $r_2 > r_1$ is determined only by the functional value at $\alpha(r_1)$, and the function $\beta(r)$. One may also interpret the exponential term on the RHS as the ratio of two functional values.
Moreover, we also have

$$(\frac{r_2}{r_1})^{2\beta(r_1)+n-1} \leq \frac{H(r_2)}{H(r_1)} \leq (\frac{r_2}{r_1})^{2\beta(r_2)+n-1}$$

Lemma 2 *Now we will get to the main lemma that involves the comparison of the doubling index and the frequency function, which was the main motivation behind introducing the frequency function. We will be comparing the frequency in a ball and the doubling index in a slightly larger ball.*

Let $\delta > 0$ be sufficiently small, and let $N(r) := N(B(0,r))$ for $r > 0$. There exists a constant $C = C(\delta, n)$ (i.e. depends on both δ and n) such that

$$\beta(r(1+\delta))(1-100\delta) - C \leq N(r) \leq \beta(2r(1+\delta))(1+100\delta) + C$$

uniformly for all $0 < r < \frac{1}{4}$ and all harmonic functions u defined in $2B(0,1) = B(0,2)$.

Lemma 3 *(Almost monotonicity of the doubling index - for concentric balls)*
The following lemma does not involve the usage of frequency functions. The structure of the inequality is comparable to that in Corollary 3.2, where instead of $H(r)$ we have $\sup_{B(0,r)} |u|$.

For a sufficiently small $\delta > 0$, and a harmonic function u in $B(0,2)$, there exists a constant $C = C(\delta, n) > 0$, independent of u such that

$$(\frac{r_2}{r_1})^{N(r_1)(1-\delta)-C} \leq \frac{\sup_{B(0,r_2)} |u|}{\sup_{B(0,r_1)} |u|} \leq (\frac{r_2}{r_1})^{N(r_2)(1+\delta)+C}$$

for all $r_1 \leq r_2/2 \leq \frac{1}{4}$

Lemma 4 *(Almost monotonicity of the doubling index - for non-concentric balls)*
Suppose B contains ball b and ball $2b$. That is,

$$b \subset 2b \subset B$$

Then, as you move from the smaller ball b to the larger ball B the doubling index cannot increase by more than a constant factor:

$$N_u(b) \leq C N_u(B)$$

*where u is a harmonic function in $2B$. The doubling index can increase/decrease **but not wildly**. It's variation is bounded and controlled by the constant $C = C(n) > 0$.*

5 Things I didn't understand on my first read.

- What's a harmonic function
  Resolved: Refer to "Basic Definitions"

- Meaning of notation $\{u = 0\}$ – I assume it's the set of values x in $\mathbb{R}^n$ for which the image under u is 0 where $u : \mathbb{R}^n \to \mathbb{R}$
  Resolved: Refer to "Basic Definitions"

- What's the Hausdorff measure of a set
  Resolved: Refer to "Basic Definitions"

- What does it mean for a harmonic function to be in $\mathbb{R}^n$
  Resolved: $D_u = \mathbb{R}^n$

- What is meant by Hausdorff dimension? Is it a property of a vector space like the dimension of a vector space?
  Resolved: The Hausdorff dimension s is the dimension at which the s-dimensional Hausdorff measure changes from $\infty \to 0$. Sort of like a stage in between. It is defined by

  $$\dim_H(E) = \inf\{s : \mathcal{H}^s(E) = 0\}$$

  or alternatively,

  $$\dim_H(E) = \sup\{s : \mathcal{H}^s(E) = \infty\}$$

  The nodal set is measured in dimension $n-1$ as the nodal set is expected to behave like an $(n-1)$ dimensional surface.

- What does it mean for a function to be in a ball $B \subset \mathbb{R}^n$. What does this notation mean: $\sup_B |h|$
  Resolved: $D_u = B$. $\sup_B |h|$ yields the supremum of all the values of $h(x)$ for all $x \in B$

- Why's $\mathcal{N}_h$ called the 'doubling index'? Reason for both words. I assume it quantifies how much the function has been 'scaled' or something similar to that?
  Resolved: Close. It's called the 'doubling index' as it quantifies the extent to which the harmonic function u changes from the ball B to $2B$. High doubling index indicates high variation of u.

- Notation: $\mathcal{H}^{n-1}(S)$ where S is the set. Is this the Hausdorff measure? And in the main theorem, is the set $\{u = 0\} \cap B$ just the points in $\mathbb{R}^n$ within the n-dimensional ball (think of it as 3D for convenience) such that u is 0 at those points?
  Resolved: Refer to "Basic Definitions"

- Harnack's Inequality for harmonic functions

- Homothetic copy, Homothety coefficient of a cube $Q \in \mathbb{R}^n$
  Resolved: Refer to "Basic Definitions"

- What does ∂B mean?
  Resolved: For a ball $B(x, r)$, ∂B represents the points on the boundary/surface of the ball. For a 3D ball, it is the sphere. Equivalently,

$$\partial B(x, r) = \{y \in \mathbb{R}^n : |y - x| = r\}$$

- Maximum principle
  Resolved: The maximum principle for harmonic functions states that for a non-constant harmonic function u in ball B its maximum cannot occur in the interior points of the ball.

6 Observations and connections

- zero sets of functions $f : \mathbb{R}^n \to \mathbb{R}$ are generically $(n-1)$-dimensional? Is this related to Hausdorff dimension/measure?

- In the definition of Hausdorff measure, we cover E by sets U_i satisfying

$$\text{diam}(U_i) \leq \delta.$$

  This reminded me of compactness, since compact sets have the property that every open cover has a finite subcover. Although worthy to note that the cover i.e. the sets U_i need not be open, unlike the case for compact sets.

7 Things I particularly liked

I liked the method of projecting the n-dimensional ball of radius r into a hyperplane which contains an (n-1)-dimensional ball of radius r - Easily understood by considering $n = 3$